

MOHAMMED F. SAFI

PhD, CCC/SLP, MS, BS-SLP/Aud.

RESEARCH/SCHOLARSHIP PORTFOLIO

RO. RESEARCH STATEMENT

Since Joining UAEU in 2018 and despite the heavy teaching load that I carried (19 different classes), I was research active. In fact, I enjoy research and research related activities. That resulted in 11 research publications in indexed international journals in addition to several conference presentations. Further, I have 8 manuscripts that are under review in Q1 and Q2 Scopus journals. Since joining UAEU, I collaborated with, and provided training to several PhD, MA, and BA students. Additionally, I collaborated with well-known international research institutes that resulted in a contribution to the field of neuroscience and rehabilitation.

Research Philosophy and Contribution.

I believe that the future of rehabilitation fields lies at the intersection of evidence-based practice, evaluating current clinical practices, and generating new effective treatment approaches. It is relatively easy to form clinical hypotheses, but gathering data to evaluate these hypotheses, rather, difficult, and rare. Effectively gathering evidence based clinical data requires the development of new perspectives and the utilization of new available technology, and their integration with current research programs.

My research activities focused on utilizing new available technology to gather clinical data in swallowing and motor speech abilities. In particular, I utilized two relatively new technologies that are available to researchers in communication sciences and disorders. The first is neuromuscular electrical stimulation (NMES). The second is functional Near-Infrared Spectroscopy (fNIRS).

Established Neuroscience of Education Lab. Since securing a start-up funds from UAEU, I established a neuroscience of education lab. Functional Near-Infrared Spectroscopy (fNIRS) brain-imaging is a novel technology measures changes in the brain's blood oxygen level density (BOLD) while a person is performing specific motor or cognitive tasks. fNIRS scanners are very small (the size of a desktop computer), portable, and particularly child friendly (children and adults sit normally in a comfortable chair, and babies can be studied while seated on mom's lap). Establishing a neuroscience lab not only generated cutting edge and evidence-based research but also stimulated and recruited millennial student researchers to the field of education and speech language pathology.

Neuromuscular electrical stimulation research. My research work focused on exploring the use of NMES on orofacial muscles in healthy volunteers and in individuals with orofacial weakness (Safi, et. al., 2014, 2017, 2018, 2020, 2021). My experimental/empirical research demonstrated that not only it's safe to utilize NMES in the orofacial region but also its potentially an effective treatment modality with individuals who have labial weakness, Another outcome from my

research was that submental electrical stimulation did not generate any effect on lingual muscles (Safi, et. al., 2017). This research explored new frontiers in neuroscience I took steps in providing evidence based clinical research. Another area that I focused on was evaluating tolerance and comfort levels of facial NMES in healthy volunteers. I have adopted a scale to measure NMES comfort level from the researched tools that are used with patients to measure their perception of pain levels accurately. The main purpose of this research was to establish whether facial NMES is tolerable or not (Safi, 2020). Indeed, one of my other major achievements and contribution in the lab was to collaborate with senior researchers at the NIH, USA and James Madison University on a neuroscience related research that resulted in a publication in the Neuromodulation journal (Scopus Q1 Neuroscience journal).

Further, since joining the Special Education Department at CEDU-UAEU, I made slight modification to my research agenda and interest to contribute to the development and research training of students in the department by focusing as well on communication disorders and language intervention in individuals with ASD. That led to 2 co-publications with my 2 of graduate students in addition to 4 more with 2 more students that were submitted and currently under review.

R1. PEER-EVALUATION BY EXTERNAL REVIEWERS

Please refer to the external reviewers' report

R2. IMPACT OF CANDIDATE'S RESEARCH ON HIS/HER DISCIPLINE SUCH AS CITATION AND I-INDEX AND H-INDEX

My research has been successfully translated to practical knowledge that I have shared with the school, students and the community.

R3. PRESENTATIONS (ORAL AND/OR POSTER) AT INTERNATIONAL RESEARCH CONFERENCES

- El Azab, E., **Safi, M.** June 28th, 2021. Cortical activation as a Result of Reading Stimuli for students with dyslexia and typically developing: New Insights from functional Near Infrared Spectroscopy Neuroimaging (fNIRS). UAE Graduate Students Research Conference 2021 (UAEGSRC'2021).
- Alnuaimi, M., **Safi, M.** June 12, 2021. Digital Social Stories and ASD Children. 5th Canadian International Conference on Advances in Education, Teaching & Technology 2021.
- **Safi, M. (May 25, 2021). Speaker/Moderator.** Voice Disorders & COVID-19. Emirate Society for Speech Language Pathology. Virtual Symposium.
- **Safi, M. (May 25, 2021). Speaker/Moderator.** Developmental Aphasia. Emirate Society for Speech Language Pathology. Virtual Symposium.
- **Safi, M. (April 10, 2021). Speaker.** The Effectiveness of Using Tele practice in Clinical Education: A Review from Dar Al Hekma University. 11th Emirates Otorhinolaryngology Audiology and Communication Disorders Congress. Virtual Symposium.
- **Safi, M. (Oct/2019). Speaker.** My Voice is My Power. Sharjah Learning Disabilities Conference. Sharjah Humanitarian City. Sharjah, UAE.
- **Safi, M. (Sept/2019). Keynote Speaker.** Effect of NMES Modality on Swallowing. UAE ENT & SLP Conference. UAE Medical Society. UAE.

- **Safi, M.** (July/2019). **Speaker.** Neuromuscular Electrical Stimulation effect on Facial Weakness. American Speech Language Hearing Association Conference. Chicago, USA.
- **Safi, M.** (April/2016). **Speaker.** *The Role of Different Medical Rehabilitation Specialties in the Management of Dysphagia.* College of Applied Medical Sciences Conference.
- **Safi, M.,** Wright-Harp, W., Lucker, J. & Payne, J. (April/2014). **Speaker.** *Effect of neuromuscular electrical stimulation on lingual, labial & buccal muscles.* ASHA Health Care Conference. Las Vegas, NV.
- **Safi, M.,** Wright-Harp, W., & Lucker, J. (March/2014). **Speaker.** *A Review of the Effect of neuromuscular electrical stimulation on facial and lingual muscles.* DCSHA Annual Conference. Rockville, MD.

R4. EXTERNAL AND INTERNAL RESEARCH GRANTS

Total funding received during my tenure at UAEU 2018 – 2021 is ~ 643,000.00 DHS.

- August 2021 (Submitted & Under Review). (Team of Experts Co-Coordinator). Developing a comprehensive Standardized Special Education Assessment Battery for Students 4 – 16 years in the UAE. Requested funding for a 2 years project is **1800,000.00 DHS.** Under consideration by MOE UAE.
Collaborating team:
CEDU: Dr. Ahmad Hemdan (Team of Experts Co-Coordinator), Dr. Najwa Alhosani, Prof. Osha Almuhairi, Prof. Ali Shehada, Dr. Ashraf Mustafa, and Dr. Maxwell Opuko.
CIT: Eng. Mohammed Musbah
CoS: Dr. Mohammed Seyam.

FUNDED GRANTS:

- **2021 - Current.** Team of Experts Coordinator and Research Expert – Review policy for disabilities early inclusion. Funding approved from Ministry of Education (MOE) of **36,000.00 DHS.**
Description: Reviewing Early inclusion services policies and in the Ministry of Education, UAE based on evidence-based research and international USA and Canadian standards.
- **2021 - Current.** Team of Experts Coordinator and Research Expert – Proposal preparation for a cutting-edge program to provide early detection, identification and intervention services for young children who have disabilities. Funding approved from Ministry of Education (MOE) of **102,000.00 DHS.**
- **2021 - Current.** Team of Member and Research Expert – Review and Examine Professional Special Education Services Standards. Funding approved from Ministry of Education (MOE) of **38,000.00 DHS.** Description: Setting professional standards for special education services in the Ministry of Education (UAE) in accordance with international best practices aims to raise the quality of the performance of special education teachers, improve their capabilities and skills, and ensure that they possess the competence required to practice the teaching profession in the field of special education, and to perform this trust in the required manner.
- **2021 (Completed).** Research and Clinical Expert - Scope of Practice Guidelines for Speech Therapist – Department of Health Abu Dhabi. Description: Reviewing SLP Scope of Practice in SEHA/Abu Dhabi Health Authority based on international USA and Canadian benchmarking.

- **2020-2021 (Completed).** Internal CEDU research incentive for faculty based on a submitted proposal. The research focused on Early ASD Symptoms as Reported by Mothers in the UAE. **17000.00 DHS.**
- **2020 (Completed).** Research and Clinical Expert - Developing and reviewing a license exam for Special Education Teachers (Sharjah Education Office, UAE). (Team member). Description: Developed, reviewed, and piloted a test of Special Education in Arabic and English to be used a licensing requirement for Special Education Teachers who are and will be working in Sharjah.
- **2019 – Current.** Education Neuroscience Research Lab. Using fNIRS technology to study various cortical activities in relation to educational task in healthy individuals and individuals with special needs. Awarded Funding (400,000 DHS). PhD Student Project. Comparing cortical activation during reading tasks between 2nd grade students who have are normal and those who have dyslexia.
- **2019 - 2020 (Completed).** **SURE Research Project.** Developing an Early Identification Scale for children 18 -36 months of age in the UAE Society. Awarded Funding (50,000 DHS)
- **2015 – 2018. King Saud University.** Neuromuscular electrical stimulation & orofacial muscles. Investigating the Need for a Master's Program in Speech Pathology & Audiology at King Saud University. Establishing Research Projects in the Department in collaboration with other faculty members.

UNFUNDED GRANTS:

- 2020. AUA-UAEU Joint Research. Title: Neurocognitive Underpinning of Co-morbidity between Developmental Dyslexia and Attention-deficit Hyperactivity Disorder: An Electrophysiological and Behavioral Investigation. Collaboration with Indian Institute of Technology. Proposed budget over 2 years was 316,000.00 DHS.

R5. QUANTITY AND QUALITY OF RESEARCH PUBLICATIONS, NOTING PARTICULARLY THE CONTRIBUTIONS AT THE UAEU

Published research papers since joining UAEU:

- **Safi, M.,** Martin, S., Gray, L., Ludlow, C., (Accepted August 2021). Healthy Volunteers Immediately Adapt to Submental Stimulation During Swallowing. Neuromodulation. (Q1 Scopus, PubMed and ISI Indexed)
- Opoku1, M., Nketsia, W., Mprah W., Agyei-Okyere, E., **Safi, M.** (Accepted August 2021). The Interaction between Typically Developing Students and Peers with Autism Spectrum Disorder in Regular Schools in Ghana: An Exploration using the Theory of Planned Behaviour. Front. Psychol. | doi: 10.3389/fpsyg.2021.752569. (Q2 Scopus, PubMed and ISI Indexed)
- **Safi, M.,** Al Sadrani, B., Mustafa, A. (Accepted September 2021). Using Voice-Assistant in Developing Speech and Social Skills in Children with Autism Spectrum Disorder. Disability and Rehabilitation: Assistive Technology. Submitted April 2021. (PubMed, ISI, Q3 Scopus Indexed)
- **Safi, M.F.,** Alnuaimi, M. and Sartawi, A. (2021), "Using digital social stories to improve social skills in children with autism: a pilot experimental single-subject study", Advances in Autism, Vol. ahead-of-print <https://doi.org/10.1108/AIA-02-2021-0013>. (Q3) (PubMed, ISI, Scopus Indexed).

- **Safi, M.**, Mohamoud, M. (2021). Effect of Submental Surface Neuromuscular Electrical Stimulation on Laryngeal Vestibule Opening in Healthy Volunteers at Rest and During Swallowing. Topics in Geriatric Rehabilitation. (Q3) (PubMed, ISI, Scopus Indexed).
- Masuadi, E. M., Mohamud, M. S., Alhassan, A. M., Alharbi, K. G., Hilabi, A. S., Alharbi, F. A., ... & **Safi, M. F.** (2021). Factors and Determinants of Choosing Pathology as a Future Career: Results From a Multi-Institution Study. Cureus, 13(6). (PubMed and ISI Indexed).
- Mustafa, A., **Safi, M.** (March, 2021). The effectiveness of a Training Program Based on Some Educational Strategies in Treating Reading Comprehension Difficulties Among Students of Learning Difficulties in the Fifth Grade. Journal of Special Education and Rehabilitation.
- **Safi, M.** (2020) Facial Neuromuscular Electrical Stimulation Comfort level. INDIAN JOURNAL OF OTOLARYNGOLOGY AND HEAD AND NECK SURGERY. (Q3) DOI: 10.1007/s12070-020-02173-5.

Manuscripts submitted recently and under review:

- **Safi, M.**, Hemdan, A., Opoku, M., Mustafa, A. The Impact of Health Status of Academic Staff on Attitudes towards COVID-19 Vaccination. Vaccines. Submitted August 28, 2021 (PubMed, ISI, Scopus Indexed Q1).
- **Safi, M.**, Opoku, M., Alshamsi, M., Mustafa, A. (under review). Practitioners' Perspectives on the Impact of Picture Exchange Communication System on Communication, Learning and Social Interaction of Children with Autism in UAE. International Journal of Disability, Development and Education. Submitted July 14, 2021. (PubMed, ISI, Scopus Indexed Q2).
- **Safi, M.**, Alshamsi, M., Opoku, M., Mustafa, A. (under review). Caregivers' Perception of the Effectiveness of Picture Exchange Communication System among children with autism spectrum disorder in the UAE. Disability and Rehabilitation: Assistive Technology. Submitted July 18, 2021. (PubMed, ISI, Scopus Indexed Q1).
- **Safi, M.**, Opoku, M., Alshamsi, M. (under review). Perceptions of Parents and Practitioners towards Culturally-Responsive Picture Exchange Communication System for Children with Autism in UAE. Disability and Rehabilitation. Submitted July 18, 2021. (PubMed, ISI, Scopus Indexed Q1).
- Okwan, C., Opoku, M., Nketsia, W., **Safi, M.**, Kumi, E., Mprah, W. (Under Review). Mothering Children with Visual Impairment in Ghana: Exploration of Expectations, Mental Health and Challenges. Journal of Child and Family Studies. (Q1) (PubMed, ISI, Scopus Indexed).
- Hemdan, A., Mustafa, A., **Safi, M.**, (under review). Special Education Teachers' Perceptions of Assistive Technology in the Inclusion of Students with Visual Impairments in Abu Dhabi Government Schools. Journal of Special Education Technology. Submitted September 2020. (Q2) (PubMed, ISI, Scopus Indexed).
- Abdat, R., Sartawi, A., **Safi, M.** (under review) Family Outcomes in Early Intervention for Children with SEND: UAE Parental Perspective. Journal of Research in Special Educational Needs. Submitted Feb 2021. (PubMed, ISI, Scopus Indexed)

R6. SUCCESSFUL SUPERVISION OF MASTER'S OR DOCTORAL STUDENTS

Current students under my supervision:

PhD Level Dissertation:

- 2020 – Current. Elazab., Elazab. Cortical activation using reading tasks in children with Dyslexia using fNIRS. Expected to defend November 2021.

Maser's Level Research Projects:

- 2021 - Current. Mariam al Shamsi. Teachers, Specialists, and Parents Perspectives Regarding the effectiveness of PECS for Children with Autism in the classroom in the UAE. Expected to defend October 2021.
- 2021 - Current. Alia Albedwawi. The perceptions of special education teachers about the effectiveness of educational games on the performance of children with Down syndrome in teaching the concepts or principles of mathematics. Expected to defend November 2021.

Completed Graduate Research by Supervisees

- Dalia M Alzyod. April/2021. The influence of age and gender on tongue strength and tongue endurance in both healthy children and children with idiopathic speech sound disorders in the UAE using the (IOPI®). MA Level.
- Aysha Al Kaabi. April/2021. Language Stimulation Activities for Children 0 - 36 Months. MA Level.
- Maitha Saeed Alnuaimi. December/2020. Digital Social Stories Improve Social Skills in Two Children with Autism: An Experimental Single- Subject Study. MA Level.

Served as internal examiner for the following research defenses

PhD Level

- Hanan Shaher Al Marashdi. September 2019. The impact of an enrichment program based on problem solving and reasoning to improve mathematical literacy of 10th grade students in the UAE in light of the Program for International Student Assessment (PISA).

Master's Level

- Amna Salem Ali Abdulla Saeed Al Yahyaei. April/2021. The impact of parent's involvement on their children with disabilities academic achievement and Social COMPETENCE IN the UAE. MA Level.
- Sanaa Hamid Rushwan. December/2020. Examining Abu Dhabi Private School Administrators' Attitudes toward Inclusive Education. MA Level.
- Fatima Alkaabi. April 16th, 2020. Challenges facing programs for discovering and nurturing gifted students from the point of view of teachers of special education in the United Arab Emirates.

R7. ANY OTHER ACHIEVEMENTS IN SCHOLARSHIP

- Developing a comprehensive proposal to establish a master's clinical program in Speech Language Pathology
- Coordinating the development of a PhD in rehabilitation sciences.
- External reviews for journals
- International Research Collaboration with India Institute of Technology
- International Research Collaboration with NIH USA & James Madison University USA
- Regional/local collaboration